

# SmartPantry AI: Intelligent Pantry Management and Food Waste Reduction System

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**Abstract**-Food waste has become a serious issue in many households due to poor pantry management, duplicate grocery purchases, forgotten ingredients, and inefficient meal planning. Existing recipe applications often prioritize internet recipe discovery instead of helping users utilize ingredients already available in their homes. This project proposes SmartPantry AI, an intelligent pantry management and meal recommendation system designed to reduce food waste and improve grocery decision-making through artificial intelligence, machine learning, and dashboard analytics. The system tracks pantry inventory, monitors expiration dates, categorizes ingredients, predicts food waste risk, and generates practical meal recommendations based on pantry-first cooking principles. SmartPantry AI integrates supervised machine learning models, recommendation systems, and predictive analytics into a centralized pantry intelligence platform. The proposed research investigates whether an AI-powered pantry management system can reduce food waste and unnecessary grocery purchases by improving pantry awareness and ingredient utilization.

**Keywords**-*Artificial Intelligence, Machine Learning, Recommendation Systems, Food Waste Reduction, Pantry Management, Predictive Analytics, Dashboard Analytics, Inventory Management*

## I. INTRODUCTION

Food waste has become a serious issue in many households, especially as grocery costs continue to increase and consumers purchase more food than they actually use. According to the United Nations Environment Programme (UNEP), large amounts of food are wasted every year due to poor inventory management, over-purchasing, forgotten ingredients, and inefficient meal planning [1]. Many households repeatedly buy duplicate groceries simply because they forget what is already available in their pantry or freezer. Existing recipe and grocery applications usually focus on recommending internet recipes instead of helping users make better use of the ingredients they already own. As a result, ingredients are often forgotten, expire unused, and contribute to unnecessary financial waste.

The purpose of this project is to design and develop SmartPantry AI, an intelligent pantry management and meal recommendation system that helps users track pantry inventory, monitor expiration dates, reduce food waste, and generate practical meal recommendations using ingredients already available in the pantry. Instead of functioning like a traditional recipe application, SmartPantry AI is designed to act as a pantry intelligence platform that helps users make smarter grocery and cooking decisions. The system combines artificial intelligence concepts, machine learning models, predictive analytics, recommendation systems, and dashboard visualization into one centralized application.

The primary research question for this project is:

“Can an AI-powered pantry management and recommendation system reduce household food waste and unnecessary grocery purchases by improving pantry awareness and ingredient utilization?”

The hypothesis for this research is:

“Users who utilize SmartPantry AI will reduce food waste and unnecessary grocery purchases by improving pantry awareness and increasing the usage of existing pantry ingredients through intelligent recommendation systems and predictive analytics.”

This project is important because food waste impacts both financial stability and environmental sustainability. Many people continue purchasing groceries without realizing similar ingredients already exist in their pantry, refrigerator, or freezer. SmartPantry AI encourages pantry-first cooking by helping users identify realistic meals that can be prepared immediately using existing ingredients instead of requiring additional shopping. The intended users include households, college students, families, and individuals who struggle with grocery organization,

duplicate purchases, forgotten ingredients, and expiration management.

The project also demonstrates practical applications of artificial intelligence, supervised machine learning, recommendation systems, predictive analytics, and human-computer interaction within a real-world consumer-focused environment [6].

## **II. BACKGROUND AND RELATED WORK**

### **A. Food Waste Reduction Systems**

Research involving food waste reduction, recommendation systems, and predictive analytics continues to grow due to increasing concerns surrounding sustainability and consumer spending. According to the Food and Agriculture Organization (FAO), large quantities of food are wasted globally every year because of poor inventory management and consumer behavior [2]. Existing household pantry systems often provide expiration reminders or grocery lists but lack intelligent recommendation systems capable of optimizing ingredient utilization.

Many current food applications prioritize recipe browsing and internet meal discovery rather than pantry-first cooking. Existing systems frequently recommend recipes that require additional grocery purchases even when users already possess enough ingredients to create practical meals at home. SmartPantry AI addresses this limitation by focusing on realistic meal recommendations based on ingredients already available in the pantry. The goal is not simply to recommend recipes, but to help users maximize the value of ingredients they already own before purchasing additional groceries.

### **B. Recommendation Systems**

Recommendation systems are commonly used in streaming platforms, e-commerce systems, and social media applications [3]. Traditional recommendation systems utilize techniques such as collaborative filtering, content-based filtering, and hybrid recommendation approaches [8]. However, many food recommendation systems prioritize recipe popularity instead of pantry optimization and food waste reduction.

SmartPantry AI differs from traditional recommendation systems because it prioritizes existing pantry inventory and grocery efficiency rather than encouraging additional purchases. The recommendation engine focuses on practical pantry-first meal generation,

grocery optimization, and reducing unnecessary spending. This allows the system to function as both a recommendation platform and a pantry decision-support system.

### **C. Machine Learning and Predictive Analytics**

Machine learning techniques such as Random Forest models are commonly used for classification and predictive analytics problems [5]. Supervised learning methods are effective for recommendation scoring, predictive modeling, and intelligent classification tasks [4]. SmartPantry AI incorporates Random Forest algorithms for both recommendation scoring and food waste prediction while maintaining explainable system behavior suitable for household users and academic evaluation. The recommendation model evaluates meals using nutritional and ingredient-based features, while the waste prediction model estimates spoilage risk based on ingredient category, expiration timeline, and quantity. These predictive analytics features support intelligent pantry management and dashboard analytics.

### **D. Dashboard Analytics and Human-Computer Interaction**

Modern analytical systems frequently utilize dashboard visualization to improve user awareness and decision-making. Human-computer interaction research emphasizes usability, cognitive simplicity, and efficient visual communication for decision-support systems. SmartPantry AI incorporates dashboard analytics including pantry health scores, ingredient categorization charts, expiration alerts, and grocery optimization suggestions to improve pantry awareness and usability.

The project integrates user-centered interface design principles alongside machine learning and recommendation systems to create a practical intelligent pantry management platform.

## **III. METHODOLOGY**

### **A. System Design**

SmartPantry AI is being developed as a web-based pantry intelligence application using Python and the Streamlit framework for the dashboard interface and visualization system. The application stores pantry inventory information including ingredient names, quantities, expiration dates, and ingredient categories within a centralized database system. The overall purpose of the

platform is to help users better understand what is already available in their pantry before purchasing additional groceries, while also reducing food waste through intelligent recommendation and analytics features.

The system architecture contains several major components:

- Pantry Inventory Management Module
- Recommendation Engine
- Dashboard Analytics System
- Grocery Suggestion Engine
- Waste Prediction Machine Learning Model
- Nutrition and Ingredient Categorization Module
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The pantry inventory module allows users to manually add, update, and manage pantry ingredients. Ingredients are automatically categorized into groups such as protein, dairy, grains, vegetables, fruits, and other categories using ingredient keyword matching and categorization logic. This allows the dashboard to visually analyze pantry composition and provide category-based analytics.

The recommendation engine analyzes pantry ingredients and compares them against structured meal templates and recipe datasets. Unlike traditional recipe applications that primarily recommend internet recipes requiring additional shopping, SmartPantry AI focuses on pantry-first meal recommendations using ingredients already available within the household. Recommendation scoring is based on ingredient matching, pantry compatibility, nutritional information, and overall meal practicality.

The dashboard analytics module provides:

- pantry health scores
- expiration alerts
- ingredient composition charts
- grocery optimization suggestions
- recommendation summaries
- pantry category visualization

The dashboard also incorporates visual analytics such as pie charts, metric cards, and recommendation panels to improve pantry visibility and simplify user interaction with pantry data. The overall objective of the system is to improve pantry awareness, reduce duplicate grocery purchases, reduce food waste, and help users make more efficient grocery and meal-planning decisions using existing pantry inventory.

## B. Recommendation Model

The recommendation component utilizes a supervised machine learning approach using a Random

Forest Regressor trained on recipe nutrition datasets and ingredient-based features obtained from publicly available food and recipe datasets [11], [14], [15].

The recommendation model evaluates meals using several nutritional and ingredient-related features including:

- calories
- protein
- carbohydrates
- fat
- ingredient count

The model is designed to generate recommendation scoring values that help prioritize realistic and nutritionally balanced meal suggestions within the SmartPantry AI system. Feature extraction and preprocessing techniques are used to structure recipe and nutrition data before training the model.

The recommendation engine combines both rule-based pantry matching logic and machine learning scoring techniques to improve recommendation quality while still maintaining explainable system behavior suitable for household users. This hybrid recommendation approach allows the system to recommend practical meals while also supporting intelligent recommendation analysis.

## C. Waste Prediction Model

The waste prediction component utilizes a Random Forest Classification model designed to estimate ingredient spoilage and food waste risk based on pantry inventory characteristics.

The model analyzes features including:

- ingredient category
- quantity
- expiration timeline
- days remaining before expiration

The waste prediction model classifies pantry items into:

- high waste risk
- medium waste risk
- low waste risk

These predictions support expiration monitoring, pantry health scoring, dashboard alerts, and food waste reduction analytics within the SmartPantry AI system.

The purpose of the waste prediction model is to identify ingredients likely to expire before usage and prioritize those ingredients within recommendation workflows and dashboard analytics. This allows the system to help users make better decisions regarding ingredient usage before spoilage occurs.

#### D. Machine Learning Training, Testing, and Validation

The machine learning models are trained, tested, and validated using supervised learning methodologies implemented through the Scikit-learn machine learning framework [10].

The recommendation model utilizes:

- train/test split validation
- Random Forest regression
- feature preprocessing
- Mean Absolute Error (MAE) evaluation metrics

The waste prediction model utilizes:

- train/test split validation
- Random Forest classification
- accuracy scoring
- classification report analysis

The datasets are divided into training and testing subsets to evaluate model generalization and predictive performance. The recommendation model uses regression evaluation metrics to measure prediction accuracy, while the waste prediction model uses classification-based performance metrics to evaluate spoilage prediction accuracy.

The machine learning models are integrated into the SmartPantry AI architecture to support:

- recommendation intelligence
- pantry analytics
- grocery optimization
- food waste prediction
- expiration monitoring

These machine learning components provide the core artificial intelligence functionality supporting the SmartPantry AI platform while also allowing the project to remain scalable for future improvements and expanded research analysis.

#### IV. RESEARCH METHODS

The project will utilize both software testing and user evaluation methods to assess system effectiveness. Functional software testing will validate pantry operations, recommendation accuracy, dashboard analytics, and machine learning prediction performance.

User testing will evaluate whether SmartPantry AI improves:

- pantry awareness

- ingredient utilization
- grocery decision-making
- food waste reduction
- pantry organization efficiency

*See Table 1*

Potential analyses for this project include:

- recommendation accuracy analysis
- machine learning performance evaluation
- user satisfaction surveys
- pantry utilization analysis
- grocery optimization analysis
- dashboard interaction analytics

Additional usability evaluation methods may include validated workload and usability measurements to evaluate interface effectiveness and cognitive simplicity.

#### V. CONCLUSION

SmartPantry AI is designed to function as an intelligent pantry management and food waste reduction platform that combines recommendation systems, machine learning, dashboard analytics, predictive modeling, and grocery optimization into a unified consumer application. The system prioritizes pantry-first cooking, expiration awareness, grocery efficiency, and practical meal planning while generating meal recommendations based on ingredients already available in the pantry.

The project demonstrates real-world applications of artificial intelligence, supervised machine learning, recommendation systems, predictive analytics, and dashboard analytics within a household management environment. SmartPantry AI also provides a scalable foundation for future improvements including dynamic recommendation intelligence, advanced pantry analytics, behavioral grocery prediction systems, and expanded user testing.

TABLE I. RESEARCH EVALUATION STRUCTURE

| Research Problem                                 | Hypothesis                                                 | Testing Method                   | Planned Analysis                                               | Expected Outcome                           |
|--------------------------------------------------|------------------------------------------------------------|----------------------------------|----------------------------------------------------------------|--------------------------------------------|
| Users waste food due to poor pantry awareness    | SmartPantry AI reduces ingredient waste                    | User testing and pantry tracking | Compare ingredient expiration frequency before and after usage | Reduced food waste                         |
| Users purchase duplicate groceries               | SmartPantry AI improves grocery awareness                  | Grocery tracking and surveys     | Analyze duplicate grocery purchases                            | Reduced duplicate purchases                |
| Users struggle with meal planning                | Recommendation engine improves meal preparation efficiency | User usability surveys           | Analyze meal preparation satisfaction                          | Improved meal planning                     |
| Users ignore expiring ingredients                | Dashboard alerts improve expiration awareness              | Dashboard interaction testing    | Analyze response to expiration alerts                          | Increased ingredient usage before spoilage |
| Traditional pantry systems lack visual analytics | SmartPantry AI improves pantry visibility                  | User interaction analysis        | Compare pantry understanding before and after usage            | Improved pantry management                 |

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